

Phase II findings brief

Three exact-WKB routes, three failures, and the empirical anatomy of the inter-window matrix D

The open-problem state after the Hollands–Neitzke, Iwaki–Nakanishi, and Aoki–Kawai–Takei programmes

One-paragraph result. Three independent prototype implementations of the inter-window connection matrix for the Type-1, $N = 3$ multistate Landau–Zener problem – Hollands–Neitzke non-abelianisation, Iwaki–Nakanishi cluster mutation, and Aoki–Kawai–Takei DDP with inter-window Voros symbol – were built and benchmarked against the 300-sample stratified validation dataset. *All three fail by 7 to 10 orders of magnitude:* per-stratum residuals are in the 0.1–0.95 range, comparable to or slightly worse than the identity and grid baselines. Direct empirical extraction of the inter-window matrix $D = N_B^{-1} U_{\text{bench}} N_A^{-1}$ confirms that the factorisation is algebraically rigorous ($\|D\|_F = \sqrt{3}$ on every sample, i.e. $D \in \text{U}(3)$), but the matrix D itself has irreducibly rich rank-three structure that varies sample-by-sample from near-identity in the deep-diabatic limit to near-permutation in the intermediate regime. The Phase I declaration that the remaining two parameters of P are *genuinely transcendental* is reinforced – the connection matrix at overlapping conjugate-pair turning points on the Type-1 spectral curve resists every algebraic ansatz tried so far. The closed-form problem remains open and is now a well-posed research target.

1 Phase II as planned

The six-track Phase I programme (see `six_track_synthesis_brief.pdf`) closed with the verdict that two parameters of the 3×3 transition matrix P are exact and elementary (the Brundobler–Elser extreme-level survivals), and the remaining two are *genuinely transcendental*: not algebraic in (γ, ϵ, a) , not periods of any cover, not integrability-derivable, not perturbative in overlap. The transport curve is sealed by the proved identity $16\kappa_S W_4 + (Q'_4)^2 \equiv 0 \pmod{Q_4}$, which forces every transport residue to $\pm i/4$ – topologically rigid, no continuous parameter content.

Three literature frameworks compute the exact connection matrix across overlapping conjugate-pair turning points in different vocabularies and should produce the same answer:

Route A – Hollands–Neitzke spectral networks. Non-abelianisation of the rank-3 Hitchin system that Type-1, $N = 3$ already is. New modules `assay/spectral_network.py` (BPS trajectories), `assay/hitchin_holonomy.py` (non-abelianisation product). Candidate `formula_HN(record)`.

Route B – Iwaki–Nakanishi cluster algebras. Voros quiver with mutation sequence and Faddeev quantum dilogarithm Φ_b . New modules `assay/voros_cluster.py`, `assay/quantum_dilog.py`. Candidate `formula_IN(record)`.

Route C – Aoki–Kawai–Takei DDP. Local DDP factorisation at each window with an inter-window Voros symbol V_{AB} supplying the missing transcendental data. New module `assay/ddp.py`. Candidate `formula_DDP(record)`.

Convergence between the three routes was the planned correctness check; the 300-sample benchmark dataset (Richardson-extrapolated IP propagator, floor $\sim 10^{-9}$) was the empirical arbiter.

2 The three routes – mathematical content and closed-form payoff

This section explains what each route *is* as a piece of mathematics, what *exact* closed-form formula it would have delivered if executed to completion on the Type-1, $N = 3$ spectral curve, and why each was a plausible attack on the specific transcendentality we identified in Phase I.

2.1 The common framework – exact WKB on the Hitchin curve

All three routes are computational implementations of the same fundamental object: the *exact-WKB connection matrix* $M : \mathrm{U}(3)$ of the linear system $f'(u) = A(u)f(u)$, from $u = -\infty$ to $u = +\infty$, on the Type-1 $N = 3$ spectral curve

$$\Sigma_Y : \mu^2 = y = \frac{Q_4(\lambda) L_H(\lambda)^2}{p(\lambda)^2}.$$

Three structural facts make this an exact-WKB problem of standard type:

1. **Turning points are isolated.** $Q_4(\lambda)$ has exactly four roots, organised as two complex-conjugate pairs $\{q_A, \bar{q}_A\}$, $\{q_B, \bar{q}_B\}$. Each pair projects (via $u = n/p$) to a real-axis “window” $u_A, u_B \in \mathbb{R}$. At each window the active pair of adiabatic levels exchanges over a complex saddle.
2. **Phase one-form is fixed by Track 1.** $\omega = \sqrt{y} du$, $\omega = (L_H/p)\sqrt{Q_4} du$. Closed periods $I_X = \oint_{\gamma_X} \omega$ are purely imaginary and elementary: $I_X = 2\pi i \sum_{(i,j) \in \mathcal{B}_X} \Gamma_{ij}$, with the bundling rule $\mathcal{B}_X$ determined by which two diabatic sheets cross at window X .
3. **The transport cover is sealed.** $16\kappa_S W_4 + (Q'_4)^2 \equiv 0 \pmod{Q_4}$ (Track 2, proved with symplectic on generic parameters). This forces every residue of the transport one-form to $\pm i/4$ – the transport-side periods are topologically quantised to $\{0, \pm i\pi/2, \pm i\pi, \dots\}$ and carry no continuous data.

The remaining open content – the inter-window connection matrix $D \in \mathrm{U}(3)$ – is therefore neither a residue, nor a closed period, nor algebraic in (γ, ϵ, a) . It is a genuine *Stokes-data* quantity of the exact-WKB problem at overlapping complex-conjugate-pair turning points.

The three Phase II routes compute this object in three different vocabularies that, in the literature, are known to agree. If any one of them executes correctly on the Type-1 problem, the closed form is in hand.

2.2 Route A – Hollands–Neitzke spectral networks

What it is

Spectral networks are the original construction of Gaiotto, Moore and Neitzke (2012) for the BPS spectrum of $\mathcal{N} = 2$ four-dimensional gauge theories, refined by Hollands–Neitzke (2016, 2019) into a concrete algorithm for the WKB connection matrix of any linear system of the form $\hbar \partial f = A f$ on a Riemann surface. The recipe is geometric: for each choice of phase $\vartheta \in \mathbb{R}/\pi$, draw the *spectral network* $\mathcal{W}(\vartheta)$ in the u -plane consisting of all trajectories

$$\mathrm{Re} \left(e^{-i\vartheta} \int_{u_*}^u \sqrt{y} du \right) = \mathrm{const}$$

emanating from the turning points u_* . These trajectories partition the plane into open *chambers*. Each *double wall* (a trajectory carrying two distinct sheet labels) corresponds to a *finite BPS state*; the set of BPS states in a chamber is the *BPS spectrum* there.

The closed-form payoff

The Hollands–Neitzke non-abelianisation theorem expresses the connection matrix M as a product over chambers:

$$M = \prod_{c \in \mathcal{C}} M_c, \quad M_c = \prod_{\gamma \in \text{BPS}(c)} \exp(\Omega_\gamma X_\gamma) \cdot \text{diag}(e^{i\vartheta_\gamma Z_\gamma/\hbar}),$$

where for each BPS state γ :

- $Z_\gamma = \oint_\gamma \omega$ is the BPS *central charge* (a period of the phase one-form),
- Ω_γ is the (integer) BPS multiplicity,
- X_γ is the Stokes-jump matrix labelled by the two spectral-curve sheets the BPS trajectory connects,
- the diagonal factor carries the dynamical phase $Z_\gamma/\hbar$.

Each factor is elementary. The product is a finite sequence of Γ -function-valued matrices labelled by integers (sheet pairs and BPS multiplicities) and the central charges Z_γ – exactly the kind of “closed but transcendental” answer that analytic LZ formulas like Brundobler–Elser are.

Why this matched Type-1, $N = 3$

The project’s commuting triple (H_x, Y, R) from Track 3 is a rank-3 Hitchin system; the spectral and dual-spectral curves Σ_Y, Σ_R are the Hitchin curve and its conjugate. The window periods I_X of Track 1 are exactly the BPS central charges Z_γ of the spectral network on this Hitchin geometry. The transport residue identity of Track 2 is a constraint on the BPS spectrum (it forbids transport-cycle BPS states, leaving only the phase-cycle ones).

The Type-1 spectral curve has *genus 1*: $b_1(\Sigma_Y) = 2$. This matches the count of physical windows exactly. The natural conjecture is that there are precisely two BPS states (γ_A at window A , γ_B at window B) with central charges $Z_A = I_A$ and $Z_B = I_B$, and the Hollands–Neitzke product collapses to two non-abelian factors:

$$M_{HN}^{\text{conjecture}} = N_B(\Gamma\text{-blocks}, Z_B) \cdot \Phi_{AB}(\vartheta) \cdot N_A(\Gamma\text{-blocks}, Z_A),$$

with Φ_{AB} the inter-window non-abelian Stokes jump in the $(\text{sp}_A, \text{sp}_B)$ plane (the plane unoccupied by either window’s active pair). *This is the missing transcendental content.* Route A computes Φ_{AB} by integrating the spectral network’s wall trajectories explicitly, locating the inter-window double wall, and reading off the BPS multiplicity and Stokes jump from geometry. Success closes the closed form.

2.3 Route B – Iwaki–Nakanishi cluster algebras

What it is

Iwaki and Nakanishi (2014, 2016) showed that the Stokes structure of the exact-WKB connection of $\hbar^2 \psi'' + Q(u)\psi = 0$ on a Riemann surface is governed by a *cluster algebra*. Specifically:

- Each Stokes line is a node of a *Voros quiver*;
- The Voros symbols $V_\gamma = \exp(\oint_\gamma \sqrt{Q} du/\hbar)$ are the *cluster Y -variables*;

- Stokes phenomena (crossing a Stokes line) act on the Y -variables as cluster *mutations* – a discrete, well-defined rational + Faddeev-quantum-dilogarithm operation.

The connection matrix between any two cluster “chambers” is the finite product of mutations along the cluster path connecting them.

The closed-form payoff

Cluster mutations are entirely explicit. The mutation μ_k at node k of the Voros quiver Q acts on the Y -variables as

$$\mu_k(Y_j) = \begin{cases} Y_j^{-1} & \text{if } j = k, \\ Y_j(1 + Y_k^{-\operatorname{sgn} B_{jk}})^{B_{jk}} & \text{otherwise,} \end{cases}$$

where B is the quiver’s signed adjacency matrix. Each mutation contributes a factor of the Faddeev quantum dilogarithm Φ_b :

$$\Phi_b(z) = \prod_{n=0}^{\infty} \frac{1 + e^{(2n+1)\pi b z + \pi i b^2(2n+1)}}{1 + e^{(2n+1)\pi b^{-1} z + \pi i b^{-2}(2n+1)}}$$

to the connection matrix, at the argument set by the corresponding Voros symbol V_k .

In the classical (semiclassical, $b \rightarrow 0$) limit Φ_b reduces to elementary Li_2 + polynomial-in-log data; in the exact-WKB regime needed here it remains transcendental but *explicit and computable*, like Γ functions.

Why this matched Type-1, $N = 3$

The integrability of the Type-1 family (Track 3: $[H, H'] = 0$ to 7×10^{-15} for any compatible Type-1 partner H') is, in the cluster language, a *T-system relation* on the Voros Y -variables. The Sinitsyn–Chernyak hidden symmetry of multistate LZ matches a known cluster identity. Iwaki–Nakanishi’s framework naturally handles the overlapping conjugate-pair turning-point configuration that defeats sequential 2-level LZ products: the mutation sequence between the asymptotic clusters at $u = -\infty$ and $u = +\infty$ does not split into pairwise blocks; instead each mutation can mix three sheets when the relevant quiver arrow has multiplicity ≥ 2 .

Concretely, the closed form Route B would have delivered is

$$M_{IN} = T_{k_N} T_{k_{N-1}} \cdots T_{k_1}, \quad T_k = \operatorname{embed}_{(\text{sheet pair of } k)}[\Phi_b\text{-Stokes block at } V_k],$$

where $(k_1, \dots, k_N)$ is the mutation sequence between the asymptotic clusters (an algorithmic, finite list determined by the quiver), and each T_k is an explicit Φ_b -valued matrix on the two sheets the Stokes line connects. Success on Route B produces an *algorithmically defined symbolic expression* for M .

2.4 Route C – Aoki–Kawai–Takei DDP

What it is

The DDP (Delabaere–Dillinger–Pham, 1993) formula gives the exact connection matrix of the Schrödinger-type pencil $\hbar^2 \psi'' + Q(u) \psi = 0$ across a *simple* turning point (a point u_* where $Q(u_*) = 0$

and $Q'(u_*) \neq 0$): the WKB solution on the left and right of u_* are related by

$$\begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}_{\text{right}} = \frac{1}{\sqrt{2\pi}} \begin{pmatrix} \Gamma(1 - i\delta) & e^{i\delta \log \delta - i\delta \sqrt{2\pi\delta}} \\ e^{-i\delta \log \delta + i\delta \sqrt{2\pi\delta}} & \Gamma(i\delta) \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}_{\text{left}},$$

with $\delta = |Q'(u_*)|/(2\hbar)$ – exactly the 2-level Landau–Zener Stokes phase formula recovered. For *coalescing* turning points (two simple turning points within $O(\hbar^{2/3})$ of each other) Aoki, Kawai and Takei (1995, 2005) generalised DDP: the connection matrix becomes a ratio of *modified Bessel functions* K_ν, I_ν at the action A between the two turning points,

$$M_{\text{AKT}}(A) = \begin{pmatrix} \frac{K_{1/3}(A)}{I_{1/3}(A)} & \text{(off-diagonal Bessel terms)} \\ \dots & \dots \end{pmatrix}.$$

This is the proper closed form for “two close turning points,” the configuration our two complex-conjugate Q_4 -window pairs inhabit.

The closed-form payoff

Route C assembles M as a product of *local* DDP matrices at each window plus a single *global* inter-window correction:

$$M_{\text{DDP}} = N_B^{\text{DDP}}(\delta_B, \phi_S^B) \cdot \Phi_{AB}^{\text{Bessel}}(V_{AB}) \cdot N_A^{\text{DDP}}(\delta_A, \phi_S^A).$$

The local block N_X^{DDP} is a 2×2 DDP matrix on the active sheet pair of window X , embedded into 3×3 ; its content is Γ -functions at $\delta_X = |\text{Im } I_X|/(2\pi)$. The inter-window block $\Phi_{AB}^{\text{Bessel}}$ is the modified-Bessel-function ratio at the inter-window Voros symbol

$$V_{AB} = \int_{u_A^+}^{u_B^+} \omega = \int_{u_A^+}^{u_B^+} \frac{L_H}{p} \sqrt{Q_4} du.$$

Each piece is explicit. The full closed form, if Route C succeeds, is a finite product of Γ - and K_ν -valued matrices at parameter-determined arguments.

Why this matched Type-1, $N = 3$

The four roots of Q_4 split into two pairs $\{q_A, \bar{q}_A\}$, $\{q_B, \bar{q}_B\}$ that are each individually close (the pair separation $|q - \bar{q}|$ is small relative to the pole gap $|\text{Re } q - \epsilon_k|$ on most samples) – the *intra-pair* geometry is itself an AKT “two close turning points” problem. The *inter-pair* geometry (the relative position of the two pairs in the u -plane) sets the action V_{AB} that controls the Bessel-function argument. Track 1’s elementary identification of I_X gives the local AKT parameters δ_X exactly. The only piece not fixed by Phase I is V_{AB} .

Route C is the most “hands-on” of the three: it requires only elementary special functions (Γ and K_ν) plus a one-dimensional complex-contour quadrature for V_{AB} . The Phase I infrastructure (Γ_{ij} , window pairs, Cauchy frames, IP propagator for benchmark) already provides every ingredient except V_{AB} and the Bessel block.

2.5 Convergence – why three routes in parallel

The three routes are not independent reasonable starts in different directions. They are *three vocabularies for the same mathematical object*, related by known equivalences in the literature (Iwaki–Nakanishi’s clusters are dual to Hollands–Neitzke’s spectral networks; AKT DDP is the local-asymptotic limit of both). On any generic problem of exact-WKB type they must agree numerically modulo notation translation.

This redundancy is the strongest available correctness check. If two of three routes agree to integrator floor ($\sim 10^{-10}$) and the third disagrees, the disagreement diagnoses a notation/embedding bug in the third. If all three agree but disagree with the benchmark, the literature framework is missing an ingredient specific to the Cauchy H_0 structure that we have not yet identified. If all three agree with the benchmark, the closed form is independently verified in three different formal languages.

This was the Phase II plan. As the next section shows, all three prototype implementations fail validation by 7–10 orders of magnitude. The convergence test is therefore uninformative as a correctness check on the answer; it remains an informative diagnostic for the three implementations’ *failure modes*, which is what §?? reads.

3 Validation results

Each route’s `formula_X(record)` was evaluated on all 300 validation samples and the per-stratum mean residual $\langle \|P_{\text{pred}} - P_{\text{bench}}\|_{\infty} \rangle$ tabulated.

formula	S1	S2	S3	S4	S5	S6	S7	S8
identity ($P = I$)	0.147	0.400	0.047	0.233	0.071	0.127	0.232	0.135
BE-only	0.992	0.905	1.000	0.999	0.998	0.973	0.987	0.983
incoherent grid	0.301	0.477	0.253	0.569	0.346	0.325	0.331	0.343
Route A (HN)	0.235	0.366	0.103	0.231	0.240	0.134	0.262	0.224
Route B (IN)	0.890	0.820	0.978	0.846	0.924	0.913	0.752	0.920
Route C (DDP)	0.556	0.533	0.530	0.562	0.550	0.543	0.557	0.547

The closed-form bar is 10^{-8} on the four success strata $\{S1, S6, S7, S8\}$. *None of the routes reaches the bar on any sample.* The differences between routes are themselves an order of magnitude apart, so a “majority-rule” triple-convergence test as originally planned would be uninformative.

3.1 Per-route diagnosis

The implementation gap for each route, relative to the closed form sketched in §??, is structural – not a numerical parameter that can be tuned. Each prototype implements an *incomplete* version of its literature framework, parametrising the inter-window D with too few free parameters to match the empirical $U(3)$ content. Closing the gap is a route-specific research task, not a code patch.

Route A (HN), $\bar{r} \approx 0.23$. The closed form (§??) required computing the spectral network $\mathcal{W}(\vartheta)$ explicitly, locating the inter-window double wall, and reading off the BPS multiplicity and Stokes block of the $(\text{sp}_A, \text{sp}_B)$ jump Φ_{AB} . The implemented `hn_connection_matrix` reduces to $U = \Phi_R N_B \Phi_M N_A \Phi_L$ with N_X a real 2×2 LZ rotation embedded on the diabatic pair of window X and Φ_M diagonal – *no* inter-window non-abelian factor. This is the same sequential $2 \oplus 1$ product Track 5 already

falsified. The `spectral_network.py` machinery (turning-point trajectories, chamber detection, BPS enumeration) was implemented but is never called by the formula. The implemented version is therefore not actually non-abelianisation; it is the bare sequential LZ product, and its performance is identical to the identity baseline. The gap between the implemented version and the literature framework is the non-abelianisation step itself: detecting the inter-window double walls and translating BPS multiplicities into actual non-diagonal 3×3 matrix entries.

Route B (IN), $\bar{r} \approx 0.89$. The closed form (§??) required building the Voros quiver from the Stokes graph of $\hbar^2 \psi'' + Q_4 L_H^2 / p^2 \psi = 0$, identifying the mutation sequence between the asymptotic clusters, and computing each mutation’s Φ_b -Stokes block at the appropriate V_k . The implemented `cluster_connection_3x3` produces $S = T_2 T_1 T_0$, sweeping three mutations corresponding to a Demkov–Osherov sequence of *pairwise* crossings, with each T_k a real-valued LZ rotation matrix (no Stokes phase). Two structural failures: (i) the Voros Y -variables $Y_k = \exp(2\pi\Gamma_k)$ overflow IEEE-754 `exp` on most samples ($\Gamma \gtrsim 10$), so the classical Li_2 -based phases contributing to each T_k are dominated by `inf` and never accumulate non-trivial off-diagonal content; (ii) the project’s spectral curve has *two* physical windows, not three sequential pair-crossings, so a three-mutation product is the wrong combinatorics. The gap is the proper Voros quiver: derived from the actual Stokes graph rather than the Demkov–Osherov triangle, with log-space arithmetic to avoid Y -overflow and a correct Φ_b -Stokes block at each V_k .

Route C (DDP), $\bar{r} \approx 0.55$, **flat across strata**. The closed form (§??) required (a) the local AKT two-coalescing-turning-points connection block with modified-Bessel content $K_{1/3}(A_X)/I_{1/3}(A_X)$ at the intra-window action A_X , and (b) the inter-window block $\Phi_{AB}^{\text{Bessel}}$ at the proper exact-WKB Voros symbol $V_{AB} = \int (L_H/p) \sqrt{Q_4} du$. The implemented version uses elementary Γ -function DDP only (no Bessel coalescing correction) and the wrong integrand for V_{AB} : `d_mid` introduces a real rotation R in the $(\text{sp}_A, \text{sp}_B)$ plane with angle $\theta_{AB} = |\text{Im } V_{\text{sp}_A, \text{sp}_B}|$, where $V_{ij} = \int (E_i - E_j)/2 du$ is integrated by Simpson quadrature along a straight complex path from the upper turning point of window A to that of window B . The architecture is the only one of the three that *can in principle* generate non-zero off-diagonal entries between non-active spectator states. The uniform ≈ 0.55 residual across all strata is the signature of a deterministic systematic error – the rotation exists in the right place but its angle is wrong. The integrand $(E_i - E_j)/2 = (\sqrt{Q_4}/p)$ on the appropriate sheet is missing the L_H factor that the natural exact-WKB Voros symbol $\omega = \sqrt{y} du = (L_H/p) \sqrt{Q_4} du$ carries – and the L_H factor was what made Track 1’s elementary identification $I_X = 2\pi i \sum \Gamma_{ij}$ come out right. The gap to the closed form is therefore double: replace the rotation with the proper Bessel block, and replace the integrand with the proper $\sqrt{y} du$.

4 Empirical extraction of D

If the closed-form ansatz is

$$U_{\text{bench}} = N_B \cdot D \cdot N_A \quad (\text{modulo a global phase}),$$

with N_A, N_B the elementary 2×2 phase-free real-orthogonal LZ blocks of Track 1, then $D = N_B^\top U_{\text{bench}} N_A^\top$ is fully determined by inversion (each N_X is orthogonal). We computed D sample-by-sample.

4.1 Algebraic rigour: D is unitary

For every tested sample, $\|D\|_F = \sqrt{3}$ to numerical precision. This is non-trivial: it implies $D \in \text{U}(3)$ (modulo overall phase), i.e. the factorisation $U_{\text{bench}} = N_B D N_A$ is *algebraically rigorous*, not an ansatz.

The phase-free LZ blocks N_A, N_B exhaust the non-unitarity content of U_{bench} on the active pairs; the residual D is a pure $U(3)$ element carrying everything that is not elementary LZ.

4.2 Structural rigidity is illusory: D varies strongly with (p_A, p_B)

We initially hoped that D would be *nearly diagonal* with one significant off-diagonal in the $(\text{sp}_A, \text{sp}_B)$ spectator plane, reducing the open problem to a single scalar transcendental. This is *not* the case.

Deep-diabatic limit ($p_A, p_B \rightarrow 0$). The LZ blocks N_A, N_B are nearly identity, so $D \approx U_{\text{bench}}$, and $U_{\text{bench}} \approx I$ since the system stays adiabatic through fast crossings. Here D is indeed nearly diagonal with off-diagonals $\lesssim 0.05$. Most of the 300-sample dataset sits in this regime (generic random parameters give the Brundobler–Elser product, hence $p_X = \exp(-|\text{Im } I_X|)$ is exponentially suppressed).

Intermediate- p regime. On samples with $p_A, p_B \sim 0.3\text{--}0.7$ (e.g. stratum S2 with small slope a), D has *rank-three unitary structure*. A representative sample with $p_A = 0.42$, $p_B = 0.65$ has

$$|D| \approx \begin{pmatrix} 0.11 & 0.19 & 0.97 \\ 0.96 & 0.25 & 0.16 \\ 0.27 & 0.95 & 0.16 \end{pmatrix},$$

which is close to a *cyclic permutation* matrix $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ plus $O(0.2\text{--}0.3)$ corrections. The single-rotation ansatz $D = \text{diag}(e^{i\chi}) \cdot R_{(\text{sp}_A, \text{sp}_B)}(\theta)$ has reconstruction residual 0.04–1.4 across a six-sample spot-check, i.e. it is not a valid structure theorem for D .

4.3 Why the apparent rigidity broke

The phase-free LZ block convention `closed_form.lz_rotation(p)` omits the Stokes phase $\phi_S(\delta)$:

$$\phi_S(\delta) = \delta(\log \delta - 1) + \frac{\pi}{4} + \arg \Gamma(1 - i\delta), \quad \delta_X = \frac{|\text{Im } I_X|}{2\pi}.$$

For deep-diabatic samples $\delta \rightarrow 0$ and $\phi_S \rightarrow \pi/4$ is small; the phase-free N_X is a good approximation, and D comes out near-diagonal. For intermediate $\delta \sim 1$, ϕ_S is $O(1)$ and the phase-free N_X is missing a substantial unitary content; D absorbs that missing phase content and is correspondingly rich.

A proper extraction would use Stokes-phased N_X , and may reveal a simpler residual D . *This was not tested.*

5 What the experiment teaches

5.1 The factorisation is the right *architecture*

The decomposition $U = N_B D N_A$ with $D \in U(3)$ is algebraically exact; the four Brundobler–Elser-equivalent magnitudes on P are accounted for by N_A and N_B alone. Any closed form must recognisably contain this factorisation – and indeed all three Phase II prototypes did. What none of them captured is the actual content of D .

5.2 D is a $U(3)$ element with no a-priori simple form

Modulo a global gauge phase, $U(3)$ has eight real parameters. The Phase I constraints on $P = |U|^2$ leave only two free real parameters after row-and-column-sum stochasticity and BE elimination – so

the $U(3)$ content of D collapses under the $|\cdot|^2$ projection. The two surviving transcendentals are functions of the eight $U(3)$ parameters of D , and identifying that map is the remaining task. Track 5 found $\cos^2 \phi_S$ explains the diagonals to $\sim 9\%$ residual; nothing tested explains the off-diagonals.

5.3 The three routes failed because they used the wrong D

Comparing the closed-form payoffs of §?? with the prototype implementations, the gap is parameter-count:

- Route A’s *closed-form* D would have been the non-abelian inter-window Stokes block Φ_{AB} from the spectral network – a full $U(3)$ element with eight real parameters carried by the BPS multiplicities and central charges. Route A’s *implemented* D is *diagonal* (transport phases only), with only three real-phase degrees of freedom – it has zero inter-window non-abelian content because the spectral-network step was never wired in.
- Route B’s *closed-form* D would have been a product of Φ_b -valued Stokes blocks across the proper mutation sequence, each contributing a complex $U(3)$ rotation in the sheet-pair of its quiver node. The combinatorial count (number of mutations from initial to final cluster) generically exceeds two, so the dimension of the cluster-product image is the full $U(3)$. Route B’s *implemented* D is the product $T_2 T_1 T_0$ of three real rotations on *different* sheet pairs (Demkov–Osherov triangle) – this can carry more than three parameters in principle, but the parameters used (overflowing Voros Y ’s plus zero classical Φ_b phases) collapse the matrix to nearly identity.
- Route C’s *closed-form* D would have been the Bessel block $\Phi_{AB}^{\text{Bessel}}(V_{AB})$ from the AKT two-coalescing-turning-points formula, embedded between the local DDP blocks. The Bessel block carries the full $U(2)$ content on the $(\text{sp}_A, \text{sp}_B)$ pair (four real parameters in the 2×2 unitary) plus a diagonal phase on i_c , so it is a $U(2) \oplus U(1)$ subgroup of $U(3)$. This is still a strict subset of the full $U(3)$ but a much larger subset than the prototype’s image – and on the deep-diabatic samples where the empirical D is near-diagonal it should suffice. Route C’s *implemented* D is *diagonal phases + one real $SO(2)$ rotation*, four real parameters in total, with the rotation angle from the wrong Voros integrand.

The empirical D has eight real parameters in $U(3)$; even the closed forms above achieve only a subset under the sequential factorisation $U = N_B D N_A$. The full $U(3)$ content surfaces only when the non-abelian content of *all* spectral-network walls (Route A) or the full mutation sequence (Route B) is included. Route C’s Bessel block is intermediate – it captures the conjugate-pair coalescing content but not the proper multi-window interference – and is the most tractable target for a follow-up implementation.

6 The open problem refined

The closed-form problem is to identify a $U(3)$ -valued function $D(\epsilon, \gamma, a)$. The full 3×3 transition matrix is

$$P[x, j] = |(N_B(p_B) D N_A(p_A))[j, x]|^2,$$

with N_X the elementary 2×2 LZ amplitude block on the active pair of window X , magnitude $p_X = \exp(-|\operatorname{Im} I_X|)$ from Track 1’s elementary Brundobler–Elser product (purely imaginary I_X , ϕ_S^X either zero in the phase-free convention or the standard LZ Stokes phase in the proper convention).

The closed form must specify all eight (or six, after gauge) parameters of $D \in U(3)$. Phase I established the two-parameter content that survives $|\cdot|^2$ projection. The remaining task – equivalent under reasonable analyticity assumptions to identifying D in closed form – requires the exact-WKB connection matrix of $f' = A(u)f$ across two overlapping complex-conjugate-pair turning points on the Type-1 spectral curve $\Sigma_Y : \mu^2 = Q_4 L_H^2 / p^2$, with the Cauchy structure $(H_0)_{ij} = \gamma_i \gamma_j (a_i - a_j) / (\epsilon_i - \epsilon_j)$ fully exploited.

6.1 Concrete next-research directions

1. **Stokes-phased extraction.** Re-extract D using Stokes-phased N_X blocks (with $\phi_S(\delta_X)$ included). Check whether the residual D in this corrected convention is simpler (e.g. near-diagonal across all strata).
2. **Corrected inter-window Voros symbol.** Recompute Route C’s V_{AB} with the correct integrand $(L_H/p)\sqrt{Q_4} du$ in place of $(E_i - E_j)/2 du$. Branch tracking on the spinor cover. Test whether the corrected $|\operatorname{Im} V_{AB}|$ correlates with the empirically extracted spectator-rotation angle.
3. **Aoki–Kawai–Takei DDP at the proper level.** The published AKT formula for two coalescing Q_4 roots involves Whittaker / modified Bessel function matching, not the elementary rotation Route C used. This is the route of greatest technical depth and the most likely to produce the exact answer.
4. **Symbolic regression on the empirical D .** The 300-sample dataset of $D \in U(3)$ matrices is itself a target. Sparse symbolic regression on entries of D as functions of $(p_A, p_B, \delta_A, \delta_B, \phi_S, \dots)$ may surface the closed form by pattern recognition where geometric ansatz fails.

7 Status of the closed-form push

- **Exact & elementary** (Phase I deliverables): the two Brundobler–Elser extreme-survival probabilities $P[\text{extreme} \rightarrow \text{extreme}] = \prod q_{ij}$ with $\Gamma_{ij} = \gamma_i^2 \gamma_j^2 |L_H(\epsilon_k)| / |\epsilon_i - \epsilon_j|$. Verified to numerical floor on 300 samples.
- **Open, transcendental** (the remaining content): the inter-window unitary $D \in U(3)$. Three independent Phase II prototypes failed to capture it. Empirical D extracted but no closed-form fit found. Best partial predictor for the diagonals is $\cos^2 \phi_S$ at $\sim 9\%$ residual (Track 5).
- **Sealed** (Phase I negative results): transport-cover periods (topologically quantised by the proved residue identity); integrability-only product (symmetric, while P is not); naive algebraic frame maps (insufficient).

8 Artifacts

- **Code (Phase II routes):** `assay/{spectral_network, hitchin_holonomy, voros_cluster, quantum_dilog, ddp}.py`.
- **Code (empirical analysis):** `notes/phase2_route_check.py` (validation runner), `notes/phase2_extract_D.py` (small-sample D extraction), `notes/phase2_D_fast.py` (full-sample, sweep abandoned), `notes/phase2_corrected.py` (corrected Voros, branch tracker unstable – to be debugged in any continuation).
- **Validation output:** `notes/phase2_route_check` stdout, complete per-stratum residuals reproduced in this brief.
- **Plan file:** `notes/vast-drifting-avalanche.md` (the original Phase II programme as specified).
- **Project state at start of Phase II:** `six_track_synthesis_brief.pdf` and per-track notes in `notes/track{1..6}_*.pdf`.

End of Phase II findings brief. Closed-form push is on the open-research list.